

U.G.C. NET EXAM COMPUTER SCIENCE (Solved with Explanation)

1. Infrared signals can be used for short range communication in a closed area using _____ propagation.

- (a) ground (b) sky
(c) line of sight (d) space

Ans : (c) Because, wireless communication is of three types:- (1) Radio wave, (2) Micro wave, (3) Infrared. Infrared signal are used for short range communication in a closed area using line of sight propagation.

2. A bridge has access to _____ address in the same network.

- (a) Physical (b) Network
(c) Datalink (d) Application

Ans : (a) Because, A function of a Bridge take place in the Data link layer that means it cannot read ID address so giving it access to the physical address of all stations connected to it.

3. The minimum frame length for 10 Mbps Ethernet is _____ bytes and maximum is _____ bytes.

- (a) 64 & 128 (b) 128 & 1518
(c) 1518 & 3036 (d) 64 & 1518

Ans : (d) The original ethernet IEEE 802.3 standard define the minimum ethernet frame size as 64 bytes and maximum as 1518 bytes.

Where payload size in ethernet defined the minimum 46 bytes and maximum 1500 bytes.

4. The bit rate of a signal is 3000 bps. If each signal unit carries 6 bits, the baud rate of the signal is _____.

- (a) 500 baud/sec (b) 1000 baud/sec
(c) 3000 baud/sec (d) 18000 baud/sec

Ans : (a) Because, Bit rate = Numbers of bits per unit time

Baud rate = Number of signal element or signal units per unit time .

Here, baud rate = (1/6)th bit rate

Therefore baud rate = (1/6) * 3000 = 500 baud /sec

5. Match the following :

List - I

- a. Physical layer
b. Datalink layer

List – II

- i. Allow resources to network access
ii. Move packets from one destination to other

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- | | |
|----------------------|--|
| c. Network layer | iii. Process to process message delivery |
| d. Transport layer | iv. Transmission of bit stream |
| e. Application Layer | v. Formation of frames |

Codes :

	a	b	c	d	e
(a)	iv	v	ii	iii	i
(b)	v	iv	i	ii	iii
(c)	i	iii	ii	v	iv
(d)	i	ii	iv	iii	v

Ans : (a) Because, **(a) Physical layer** : Responsible for transmission of packet in the stream of bit with proper encoding method.

(b) Data Link Layer : Formation of frame means framing is done at this layer and maximum frame size is 1500B.

(c) Network Layer : Door to door delivery of packet

(d) Transport Layer : End to end delivery

(e) Application Layer : Allow resources to Network access.

6. A grammar G is LL(1) if and only if the following conditions hold for two distinct productions $A \rightarrow \alpha \mid \beta$

- I. $\text{First}(\alpha) \cap \text{First}(\beta) \neq \{a\}$ where a is some terminal symbol of the grammar.
 - II. $\text{First}(\alpha) \cap \text{First}(\beta) \neq \lambda$
 - III. $\text{First}(\alpha) \cap \text{Follow}(A) = \phi$ if $\lambda \in \text{First}(\beta)$
- | | |
|----------------|-------------------|
| (a) I and II | (b) I and III |
| (c) II and III | (d) I, II and III |

Ans : (d) Because, Assume $S \rightarrow aX \mid aYb$

$\text{first}(aX) = a$

$\text{first}(aYb) = a$

So there will 2 entries for S in terminal 'a' which violates the rule of LL(1) grammar because for every variable and for every terminal there can be at most 1 production.

Hence their intersection should be ϕ

So, first one statement is correct

Second one is true with similar reasoning as for first.

Now, third assume example

(1) $S \rightarrow aSA \mid \epsilon$

(2) $A \rightarrow C \mid \epsilon$

Now take 2) Production

first (c) = c

follow (A) = follow (s) $\Rightarrow$ first (A) = C, E, \$

Now the first (C) $\cap$ follow (A) = C

Means there will be again 2 entries for A in C place that means again violation of LL(1) rules.

So first (C) $\cap$ follow (A) = ϕ is true

Hence the answer is (d).

7. Which of the following suffices to convert an arbitrary CFG to an LL(1) grammar?

- (a) Removing left recursion alone
- (b) Removing the grammar alone
- (c) Removing left recursion and factoring the grammar
- (d) None of the above

Ans : (d) Removing left recursion and factoring from the grammar do not suffice to convert an arbitrary CFG to LL(1) grammar.

Since, LL(1) parser is top down parser. For top-down parsers, the grammar should be unambiguous, deterministic and from left recursion.

All the conditions must be satisfied.

8. A shift reduce parser suffers from

- (a) shift reduce conflict only
- (b) reduce reduce conflict only
- (c) both shift reduce conflict and reduce reduce conflict
- (d) shift handle and reduce handle conflicts

Ans : (c) Because, A shift reduce and parses the input text in one forward pass over the text, without backing up (That forward direction is generally left to right within a line, and top to bottom for multiline inputs)

9. The context free grammar for the language $L = \{a^n b^m c^k \mid k = |n - m|, n \geq 0, m \geq 0, k \geq 0\}$ is

- (a) $S \rightarrow S_1 S_3, S_1 \rightarrow a S_1 c \mid S_2 \mid \lambda, S_2 \rightarrow a S_2 b \mid \lambda, S_3 \rightarrow a S_3 b \mid S_4 \mid \lambda, S_4 \rightarrow b S_4 c \mid \lambda$
- (b) $S \rightarrow S_1 S_3, S_1 \rightarrow a S_1 S_2 c \mid \lambda, S_2 \rightarrow a S_2 b \mid \lambda, S_3 \rightarrow a S_3 b \mid S_4 \mid \lambda, S_4 \rightarrow b S_4 c \mid \lambda$
- (c) $S \rightarrow S_1 \mid S_2, S_1 \rightarrow a S_1 S_2 c \mid \lambda, S_2 \rightarrow a S_2 b \mid \lambda, S_3 \rightarrow a S_3 b \mid S_4 \mid \lambda, S_4 \rightarrow b S_4 c \mid \lambda$
- (d) $S \rightarrow S_1 \mid S_3, S_1 \rightarrow a S_1 c \mid S_2 \mid \lambda, S_2 \rightarrow a S_2 b \mid \lambda, S_3 \rightarrow a S_3 b \mid S_4 \mid \lambda, S_4 \rightarrow b S_4 c \mid \lambda$

Ans : (d) Because, Elimination Method

(1) $S \rightarrow S_1 S_3 \rightarrow a S_1 c a S_3 b$
 $\rightarrow ac ab$

(which will not be generated by the Language.)

(2) $S \rightarrow S_1 S_3 \rightarrow a S_1 S_2 c a S_3 b$
 $\rightarrow ac ab$ (same case as above)

(3) $S \rightarrow S_1 S_2, S \rightarrow S_1 \rightarrow a S_1 S_2 c$
 $a S_1 S_2 c \rightarrow aa S_1 S_2 c S_2 c$
 $\rightarrow aa S_2 c S_2 c$
 $\rightarrow aa ca S_2 bc$
 $\rightarrow aa ca bc$

(Which cannot be generated)

Hence the answer is (d).

10. The regular grammar for the language $L = \{w | n_a(w) \text{ and } n_b(w) \text{ are both even, } w \in \{a, b\}^*\}$ is given by :

- (a) $p \rightarrow aq \mid br \mid \lambda, q \rightarrow bs \mid ap$
 $r \rightarrow as \mid bp, s \rightarrow ar \mid bq, p$ and s
are initial and final states.
- (b) $p \rightarrow aq \mid br, q \rightarrow bs \mid ap$
 $r \rightarrow as \mid bp, s \rightarrow ar \mid bq, p$ and s
are initial and final states.
- (c) $p \rightarrow aq \mid br \mid \lambda, q \rightarrow bs \mid ap$
 $r \rightarrow as \mid bp, s \rightarrow ar \mid bq$
 p is both initial and final states.
- (d) $p \rightarrow aq \mid br, q \rightarrow bs \mid ap$
 $r \rightarrow as \mid bp, s \rightarrow ar \mid bq$
 p is both initial and final states.

Ans : (c) Because, This is Even – Even language.
Number of a a's are divisible by 2 and number of b's are divisible by 2.

$$P \rightarrow aA \mid bR \mid \lambda$$

$$Q \rightarrow bS \mid aP$$

$$R \rightarrow aS \mid p$$

$$S \rightarrow aR \mid bQ$$

⇒ Design DFA for language.

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graph LR
    P((P)) -- a --> Q((Q))
    Q -- b --> P
    P -- b --> R((R))
    R -- a --> P
    R -- a --> S((S))
    S -- b --> R
    S -- b --> Q
    Q -- a --> S
    style P fill:#fff,stroke:#000
    style Q fill:#fff,stroke:#000
    style R fill:#fff,stroke:#000
    style S fill:#fff,stroke:#000
    
```

Convert DFA to CFG

Hence option (c) is correct.

11. KPA in CMM stands for
- (a) Key Process Area
 - (b) Key Product Area
 - (c) Key Principal Area
 - (d) Key Performance Area

Ans : (a) Because, Key Process Area (KPA) used in CMM identifies a cluster of related activities that, when performed collectively, achieve a set of goals considered important.

12. Which one of the following is not a risk management technique for managing the risk due to unrealistic schedules and budgets?
- (a) Detailed multi source cost and schedule estimation.
 - (b) Design cost
 - (c) Incremental development
 - (d) Information hiding

Ans : (d) Because, Risk management is the Process of identifying, controlling threats to an organization's capital and earning. Risk management can be considered as dealing with the possibility and actual occurrence of those events that are not regular expected and information hiding is the principle of segregation of the design decision in a computer program that are most like to change, thus protecting other parts of the program from extensive modification if the design decision is changed.

13. _____ of a system is the structure or structures of the system which comprise software elements, the externally visible properties of these elements and the relationship amongst them.

- (a) Software construction
- (b) Software evolution
- (c) Software architecture
- (d) Software reuse

Ans : (c) Because, Software architecture refers to the high level structure of software system, the discipline of creating such structures, and the documentation.

14. In function point analysis, the number of complexity adjustment factors is

- (a) 10
- (b) 12
- (c) 14
- (d) 20

Ans : (c) There are 14 relative complexity adjustment factor are

- (1) Requirement for reliable backup and recovery
- (2) Require for data communication
- (3) Extent of distributing processing
- (4) Performance requirement
- (5) Expected operational environment
- (6) Extent of online data entries
- (7) Extent of multi-operation online data input.
- (8) Extent of online updating of master files.
- (9) Extent of complex inputs, outputs.
- (10) Extent of complex data processing.
- (11) Extent of currently developed code can be designed for reuse.
- (12) Extent of conversion and installation in the design.
- (13) Extent of multiple installation in an organization.
- (14) Extent of change and focus on ease of use.

15. Regression testing is primarily related to

- (a) Functional testing
- (b) Development testing
- (c) Data flow testing
- (d) Maintenance testing

Ans : (d) Because, regression testing is the process of testing changes to computer program to make sure that the older programming still works with the new changes. This happens only in case of maintenance. So it relates to the maintenance testing, hence the answer is option (d).

- (a) Both I & II are true.
 (b) Both I & II are false.
 (c) I – false, II – true
 (d) I – true, II – false

Ans : (a) Because, if there exists a particular X such that $P(x)$ is true then obviously there exists some x for which $P(x)$ is true

Hence I is correct

Now $\sim \forall P(x)$ means $\exists \sim P(x)$

Therefore 2 is also correct and the right answer is option (a).

20. Give a compound proposition involving propositions p, q and r that is true when exactly two of p, q and r are true and is false otherwise.

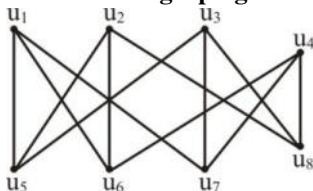
- (a) $(p \vee q \wedge \neg r) \wedge (p \wedge \neg q \wedge r) \wedge (\neg p \wedge q \wedge r)$
 (b) $(p \wedge q \wedge \neg r) \wedge (p \wedge q \wedge \neg r) \wedge (\neg p \wedge q \wedge r)$
 (c) $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \wedge (\neg p \wedge q \wedge r)$
 (d) $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$

Ans : (d) Because,

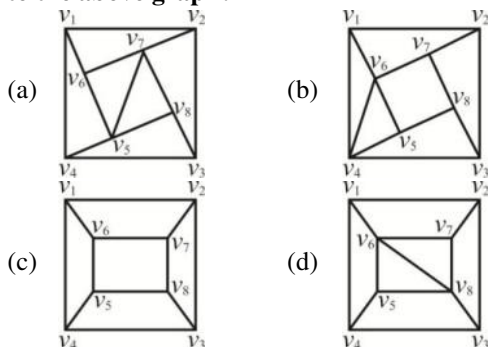
P	Q	E	Function	Option A	Option B	Option C	Option D
T	T	F	T	F			T
T	F	T	T		F		T
F	T	T	T			X	T

Hence the answer is (d).

21. Consider the graph given below as:



Which one of the following graph is isomorphic to the above graph?



Ans : (c) Because, from given figure we can say that, every vertex has degree = 3 in all options but C, graphs has at least one vertex with degree = 4, and then (c) has 12 edge which is equal to above graph edge.

22. The upper bound and lower bound for the number of leaves in a B-tree of degree K with height h is given by :

- (a) K^h and $2^{\lceil k/2 \rceil^{h-1}}$ (b) K^*h and $2^{\lfloor k/2 \rfloor^{h-1}}$
 (c) K^h and $2^{\lfloor k/2 \rfloor^{h-1}}$ (d) K^*h and $2^{\lceil k/2 \rceil^{h-1}}$

Ans : (a) Because, consider a B-tree of degree K with height h . So K^h & $2^{\lceil K/2 \rceil^{h-1}}$ means K^h and $2^{\lceil K/2 \rceil^{h-2}}$, Hence the correct answer is option (a).

23. Consider a complete bipartite graph $k_{m,n}$. For which values of m and n does this, complete graph have a Hamilton circuit

- (a) $m = 3, n = 2$ (b) $m = 2, n = 3$
 (c) $m = n \geq 2$ (d) $m = n \geq 3$

Ans : (c) Because, $K_{m,n}$ is Hamilton circuit if and only if $m = n \geq 2$, (Circuit is a path similar to a cycle that starts and ends at the same vertex) and $K_{m,n}$ is a Hamilton circuit if $m = n \geq 2$, so the correct answer is (c).

24. Big-O estimates for the factorial function and the logarithm of the factorial function i.e. $n!$ and $\log n!$ is given by

- (a) $O(n!)$ and $O(n \log n)$
 (b) $O(n^n)$ and $O(n \log n)$
 (c) $O(n!)$ and $O(\log n!)$
 (d) $O(n^n)$ and $O(\log n!)$

Ans : (b) Because, In complexity theory $n! = n^n$
 $O(n!) = O(n^n)$
 and $O(\log(n!)) = O(n \log(n))$
 So the answer is (b).

25. How many cards must be chosen from a deck to guarantee that at least

- two aces of two kinds are chosen.
 - two aces are chosen.
 - two cards of the same kind are chosen.
 - two cards of two different kinds are chosen.
- (a) 50, 50, 14, 5 (b) 51, 51, 15, 7
 (c) 52, 52, 14, 5 (d) 51, 51, 14, 5

Ans : (a) Because, since we have to be sure consider the worst cases for all

- Two aces same kind are chosen and first 48 card without ace then 49th is the type of ace and 50th also be a type of ace.
- Here also 49th and 50th are different of aces, just as above (I).
- We suppose that in the worst case we draw 4 cards of different kinds. Then the next card must match with any one of the previously draw cards. So atleast 5 cards must be drawn.
- We suppose that in the worst case we draw 13 cards of same kind. Then surely the 14th card from the remaining 39 cards will be of different kind. So total 14 cards should be drawn.

Then the answer is 50, 50, 14, 5 means the correct option is (a).

26. Match the following with respect to the mobile computing technologies :

List - I		List – II	
a. GPRS	i.	An integrated digital radio standard	
b. GSM	ii.	3G wireless/Mobile technology	
c. UMTS	iii.	Nine different schemes for modulation and error correction	
d. EDGE	iv.	An emerging wireless service that offers a mobile data	

Codes :

	a	b	c	d
(a)	iii	iv	ii	i
(b)	iv	i	ii	iii
(c)	ii	iii	iv	i
(d)	ii	i	iv	iii

Ans : (b) Because, GPRS (General Packet Radio Service). It is an emerging wireless service that offers a mobile data. It is a packet based wireless communication that complement circuit switched network.

GSM (Global System for Mobile Communication) : The technology behind the global system for mobile communication (GSMTM) uses Guassian minimum shifting keying (GMSK) modulation.

UMTS (Universal Mobile Telecommunication System): UMTS is an umbrella term for the third generation radio technologies with 3GPP. The radio technologies access specification provide for Frequency Division Duplex (FDD) and time division duplex (TDD) variant. EDGE (Enhanced GPRS or EGPRS) : It is a data system used on top of GSM networks. It provide nearly three times fastest speed than the GPRS system.

After considering all, the right answer is (b).

27. Object Request Broker (ORB) is

- I. A software program that runs on the client as well as on the application server.
 - II. A software program that runs on the client side only.
 - III. A software program that runs on the application server, where most of the components reside.
- (a) I, II & III (b) I & II
(c) II & III (d) I only

Ans : (d) Because, object request broker (ORB) is middle wave that uses the COBRA (Common Object Request Broker Architecture) specification. The ORB takes care of all of the details in routing a request from client to object and the response to its destination. ORB runs on clients and as well as server application. Hence the answer is (d).

Ans : (a) Because, secure electronic transaction (SET) protocol is used to facilitate the secure transmission of consumer credit card information via electronic avenues, such as the internet. SET blocks out the details of credit card information.

31. In a paged memory management algorithm, the hit ratio is 70%. If it takes 30 nanoseconds to search Translation Look-aside Buffer (TLB) and 100 nanoseconds (ns) to access memory, the effective memory access time is

- (a) 91 ns (b) 69 ns
(c) 200 ns (d) 160 ns

Ans : (d) Because, we can assume single level paging and TLB look and page table look happens sequentially meaning TLB look time is to be considered even for TLB miss on a TLB miss we need to access main memory for getting the physical address for page table doing the actual memory access for the data.

Effective access time = $H_i + \text{Ratio} * \text{Time during hit} + \text{Miss ratio} * \text{Time during miss}$

$$\Rightarrow 0.7 * (30 + 100) + 0.3 (30 + 100 + 100)$$

$$\Rightarrow 91 + 69$$

$$\Rightarrow 160 \text{ ns}$$

$\Rightarrow$ Hence the answer is option (d).

32. Match the following :

- | List - I | List - II |
|------------------------------|---|
| a. Multilevel feedback queue | i. Time-slicing |
| b. FCFS | ii. Criteria to move processes between queues |
| c. Shortest process next | iii. Batch |
| d. Round robin scheduling | iv. Exponential smoothening |

Codes :

- | | a | b | c | d |
|-----|-----|-----|----|----|
| (a) | i | iii | ii | iv |
| (b) | iv | iii | ii | i |
| (c) | iii | i | iv | i |
| (d) | ii | iii | iv | i |

Ans : (d) Because,

Multilevel Feedback queue: In multilevel, except jobs may be moved from one queue to another for a variety of reasons.

FCFS : It allocate the CPU in order in which the process arrive. When the CPU is free, it is allowed to process, which is occupying the front is the queue.

Shortest Process Next : The process with the shortest expected processing time is selected for execution, among the available process in the ready Queue. Thus a

short process will jump to the head of the Queue over long jobs.

Round Robin Scheduling : This type of scheduling algorithm is basically designed for time sharing system. It is similar to preemptive added. Round robin is also known as time-slicing scheduling.

⇒ After consider all the scheduling algorithm, the correct answer is option (d).

33. Consider a system with five processes P_0 through P_4 and three resource types R_1 , R_2 and R_3 . Resource type R_1 has 10 instances, R_2 has 5 instances and R_3 has 7 instances. Suppose that at time T_0 , the following snapshot of the system has been taken :

Allocation			
	R_1	R_2	R_3
P_0	0	1	0
P_1	2	0	0
P_2	3	0	2
P_3	2	1	1
P_4	0	2	2

Max		
R_1	R_2	R_3
7	5	3
3	2	2
9	0	2
2	2	2
4	3	3

Available		
R_1	R_2	R_3
3	3	2

Assume that now the process P_1 requests one additional instance of type R_1 and two instance of resource type R_3 . The state resulting after allocation will be

- (a) Ready state (b) Safe state
(c) Blocked state (d) Unsafe state

Ans : (b) Because, By taking one instance of R_1 , P_1 will fulfill its need maximum R_1 and By taking two instance of R_3 , P_1 will fulfill its need of maximum R_3 now, it needs two instances of R_2 after which it can complete execution and release all of its resources.

Hence the correct answer is option (b) safe state.

34. Match the following :

List - I		List – II	
a.	Contiguous allocation	i.	This scheme supports very large file sizes.
b.	Linked allocation	ii.	This allocation technique supports only sequential files.
c.	Indexed allocation	iii.	Number of disks required to access file is minimal.

- d. Multi-level indexed iv. This technique suffers from maximum wastage of space in storing pointers.

Codes :

	a	b	c	d
(a)	iii	iv	ii	i
(b)	iii	ii	iv	i
(c)	i	ii	iv	iii
(d)	i	iv	ii	iii

Ans : (b) Because,

(i) Contiguous Allocation : In this each file occupies a set of contiguous blocks on the disk and it have randomly accessed and a wasteful of space in this files cannot grow.

(ii) Linked Allocation : In this each file is a linked list of disk blocks and it need only starting address and it have no random accessed.

(iii) Indexed Allocation : It brings all pointer into the indexed block and it needs logical view and it needs index table and it have random access.

(iv) Multilevel Indexed : A multilevel index can be created for any type of first level index (Primary, secondary, clustering) as long as the first level index consist of more than one disk block and it supports very large files.

⇒ So after considered all the allocation's the true answer is option (b).

35. Which of the following commands will output "one two three"?

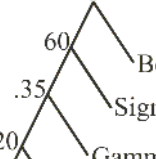
- (a) for val; do echo-n \$val; done < one two three
 (b) for one two three; do echo-n-; done
 (c) for n in one two three; do echo-n \$n; done
 (d) for n in one two three {echo -n \$ n}

Ans : (c) Because, for n in one two three; do echo-n-; done is the command that give the output "one two three" So the answer is (c).

36. Merge sort makes two recursive calls. Which statement is true after these two recursive calls finish, but before the merge step?

- (a) The array elements form a heap.
 (b) Elements in each half of the array are sorted amongst themselves.
 (c) Elements in the first half of the array are less than or equal to elements in second half of the array.
 (d) All of the above

Ans : (b) Because, merge sort is possible when two arrays are sorted. If two unsorted arrays are submitted to merge sort algorithm. It will sort each of them first before merging them, because merge sort is a divide and conquer algorithm, Hence the answer is option (b).

- Ans : (b)** Because,
- 
- Now,
- Path length with 4 = alpha, delta (.48 + .32)
- Path length with 3 = gamma (.45)
- Path length with 2 = sigma (.50)
- Path length with 1 = beta (.40)
- Average path length = $(4 \times 0.12) + (1 \times 0.40) + (3 \times 0.15) + (4 \times 0.8) + (2 \times 0.25)$
- Hence the correct answer is option (b) 2.15

- Ans : (c)** Because, In searching for an element in hash table requires $O(1)$ time for the average time, whereas for direct it holds for the worst-case time. So the correct answer is option (c).

- Ans : (b)** Because, In determining the time complexity of an algorithm, if there is a function $T(n) = n^2 - 11n - 9$, n^2 is always considered since it is of highest degree. The lower terms are ignored since they contribute most nothing to the sum when n is sufficiently large.

40. What is the maximum number of parenthesis that will appear on the stack at any one time for parenthesis expression given by

(() (()) (()))

- (a) 2 (b) 3
(c) 4 (d) 5

Ans : (b) Because,

Input	Stack
((()((()((()))	empty
()((()((()((	([PUSH]
)((()((()((	(([PUSH]
((()((()((	(→ [Pop as parenthesis match]
))((()((	(([PUSH]
))((()((	(([PUSH]
)((()((	((→ [Pop as Parenthesis match]
)))	(([PUSH]
))	(([PUSH]
)	(([POP]
)	([POP]
Empty	Empty (POP)

⇒ As you seen the maximum 3 parenthesis appear at stack at one time. So the answer is option (b).

41. Match the following :

List - I	List - II
a. Automatic storage class	i. Scope of the variable is global.
b. Register storage class	ii. Value of the variable persists between different function calls.
c. Static storage class	iii. Value stored in memory and local to the block in which the variable is defined.
d. External storage class	iv. Value stored in CPU registers.

Codes :

	a	b	c	d
(a)	iii	iv	i	ii
(b)	iii	iv	ii	i
(c)	iv	iii	ii	i
(d)	iv	iii	i	ii

Ans : (b) Because,

(a) Automatic Storage Class : In this variable Automatically created and destroyed within a function and value is stored in memory and local to the block.

(b) Register Storage Class : In this variable that is allocated one register of the machine instead of conventional memory allocation and values are stored in CPU register.

(c) Static Storage Class : In this the variable that is local to a function in which it is define and holds the value for the entire program.

(d) External Storage Class : In this variable that recognized globally and can be accessed from any function that falls within its scope.

⇒ So after considered all the storage classes, we find that the correct option is (b).

42. When we pass an array as an argument to a function, what actually gets passed?

- (a) Address of the array
- (b) Values of the elements of the array
- (c) Base address of the array
- (d) Number of elements of the array

Ans : (c) Because, Base address of an array is the starting address of an array. One can easily access to all the cells of an array if information about this can be achieved.

**43. While (87) printf ("computer");
The above C statement will**

- (a) print "computer" 87 times
- (b) print "computer" 0 times
- (c) print "computer" 1 times
- (d) print "computer" infinite times

Ans : (d) Because, Parameter of a while statement in a languages takes two valves true in the form of non-zero and false in the form of zero. Here 87 is a non-zero valve and hence the while block gets true for infinite times.

44. A friend function can be used to

- (a) avoid arguments between classes.
- (b) allow access to classes whose source code is unavailable.
- (c) allow one class to access an unrelated class.
- (d) None of the above

Ans : (c) Because, unrelated class means the classes are not connected via inheritance. But a relation can be achieved between two unrelated classes through friend function.

45. Which of the following is the correct value returned to the operating system upon the successful completion of a program?

- (a) 0
- (b) 1
- (c) - 1
- (d) Program do not return a value.

Ans : (a) Because, 0 is the identifier for a program to inform the operating system (OS) about proper running of a program. Each of the non-zero values signifies unique error during running of a program.

46. Manager's salary details are hidden from the employee. This is called as

- (a) Conceptual level data hiding
- (b) Physical level data hiding
- (c) External level data hiding
- (d) Local level data hiding

Ans : (c) Because, External level (highest level). It is the user's view of the database and it describes part of the database for Particular group of user and provides a powerful and flexible security mechanism by hiding part of the database from certain users. The user is not aware of the existence of any attribute that are missing from the view. It permits user to access data in a way that is customized to their needs. So that the same data can be seen by different user in different ways, at the same time.

47. Which of the following statements is false?

- (a) Any relation with two attributes is in BCNF.
- (b) A relation in which every key has only one attribute is in 2NF.
- (c) A prime attribute can be transitively dependent on a key in 3NF relation.
- (d) A prime attribute can be transitively dependent on a key in BCNF relation.

Ans : (d) Because, A relation is in BCNF, if and only if, every determinant is a candidate key i.e. the left hand side of every non trivial dependency is a super key or there is a trivial dependency or both. After 3NF, there may arise a situation that two candidate keys are having one attribute in common (overlapping) and this overlapping must be cleared to satisfy BCNF. A prime attribute (an attribute that is a part of a candidate key) can not depend on a super key. So a prime attribute cannot be transitively dependent on a key in BCNF relation. So the false statement is option (d).

48. A clustering index is created when ____.

- (a) primary key is declared and ordered
- (b) no key ordered
- (c) foreign key ordered
- (d) there is no key and no order

Ans : (a) Because, clustered index determines the order in which the rows of a table are stored on disk. If a table has a clustered index. Then the rows of that table will be stored on disk in the same exact order as the clustered index. In some cases, the index is created on non-primary key columns which may not be unique for each record. Clustering index is defined as an ordered data files, So the correct answer is option (a)

49. Let $R = \{A, B, C, D, E, F\}$ be a relation schema with the following dependencies $C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$
- Which of the following is a key for R ?
- (a) CD (b) EC
 (c) AE (d) AC

Ans : (b) Because,

$\{CD\}^+ = \{CDF\}$
 $\{FC\}^+ = \{ABCDEF\}$
 $\{AE\}^+ = \{ABE\}$
 $\{AC\}^+ = \{ABCF\}$

$\Rightarrow$ So the correct option is (b).

50. Match the following :

List - I		List - II	
a. DDL		i. LOCK TABLE	
b. DML		ii. COMMIT	
c. TCL		iii. Natural Difference	
d. BINARY		iv. REVOKE	
Operation			
Codes :			
a	b	c	d
(a) ii	i	iii	iv
(b) i	ii	iv	iii
(c) iii	ii	i	iv
(d) iv	i	ii	iii

Ans : (d) Because,

(a) **DDL** : Data definition language (DDL): Create, alter drop schema object. Grant and revoke privileges and roles.

(b) **DML** : Data manipulation language (DML) statement access and manipulate data in existing schema objects and the DML statements are : (ALL, DELETE, INSERT, LOCK, TABLE, MERGE etc.)

(c) **TCL** : Transaction control language (TCL) and the TCL statements are : COMMIT, ROLL BACK, SAVE POINT etc.

(d) **BINARY Operation** : Addition, difference, multiplication, division etc.

$\Rightarrow$ So the correct answer is option (d).